

S24 — Probability: Experiment, Outcome, Event & Sample Space

Instruction: Dataset/values vertical hi diye gaye hain. Hints nahi hain.

1. Experiment vs Outcome vs Event

A company randomly selects one order and checks whether it was delivered on time.

Answer:

1. What is the **experiment**?
 2. Give one possible **outcome**.
 3. Define an **event**.
 4. Is “delivered late” an outcome or an event? Explain.
 5. Give another possible event.
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2. Sample Space — Die

A fair die is rolled once.

Answer:

1. Write the complete **sample space**.
2. Number of possible outcomes.
3. Event A = getting an even number.
4. Event B = getting a number greater than 3.
5. Number of outcomes in A.
6. Number of outcomes in B.
7. Find .
8. Find .

3. Sample Space — Two Coins

Two fair coins are tossed.

Answer:

1. Write the complete sample space.
2. Number of possible outcomes.
3. Event A = exactly one Head.
4. Event B = at least one Head.
5. Find .
6. Find .
7. Which event has the higher probability?

4. Identify the Type of Event

Classify each event as **certain, impossible, or possible**.

1. Rolling a die and getting a number from 1 to 6.
2. Rolling a die and getting 8.
3. Rolling a die and getting an even number.
4. Selecting a red card from a standard deck.
5. Selecting a green card from a standard deck.
6. Selecting a card from the deck.

5. Basic Probability

A box contains the following balls:

Red

Red

Red

Blue

Blue

Green

Green

Yellow

Yellow

Yellow

One ball is randomly selected.

Answer:

1. Total number of balls.
2. Probability of Red.
3. Probability of Blue.
4. Probability of Green.
5. Probability of Yellow.
6. Probability of **not** selecting Yellow.
7. Which color has the highest probability?
8. Which color has the lowest probability?

6. Business Probability

An e-commerce company records the following customer outcomes:

Purchased

Purchased

Purchased

Purchased

Purchased

Added to Cart

Added to Cart

Added to Cart

Browsed Only

Browsed Only

One customer is randomly selected.

Answer:

1. Write the sample space.
 2. Define event A = customer purchased.
 3. Define event B = customer added to cart.
 4. Calculate .
 5. Calculate .
 6. Calculate .
 7. Calculate .
 8. Which outcome is most probable?
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7. Mutually Exclusive Events

A die is rolled once.

Consider:

- A = getting an even number
- B = getting an odd number

Answer:

1. Can A and B happen at the same time?
 2. Are A and B mutually exclusive?
 3. Find .
 4. Find .
 5. Find .
 6. Explain why your answer makes sense.
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8. Non-Mutually Exclusive Events

A standard deck of 52 cards is used.

- A = selecting a King
- B = selecting a Heart

Answer:

1. Can A and B happen at the same time?
 2. Are A and B mutually exclusive?
 3. How many Kings are there?
 4. How many Hearts are there?
 5. How many cards are both King and Heart?
 6. Find .
 7. Find .
 8. Find .
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9. Find the Wrong Interpretation

A data analyst says:

“The probability that a customer purchases is 0.70. Therefore, if exactly 10 customers are selected, exactly 7 of them will purchase.”

Answer:

1. Is the statement correct?
 2. What is wrong with the conclusion?
 3. What does probability 0.70 actually mean?
 4. Would 6 purchases out of 10 contradict the probability of 0.70? Explain.
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10. Full Data Analyst Probability Case

A food-delivery platform analyzes 200 randomly selected orders.

Order Status	Number of Orders
Delivered On Time	120
Delivered Late	50
Cancelled	30

Answer:

1. Define the experiment.
2. Write the sample space.
3. Define event A = order delivered on time.
4. Define event B = order delivered late.
5. Calculate .
6. Calculate .
7. Calculate .
8. Calculate .
9. Calculate .
10. Which outcome has the highest probability?
11. Are the three order-status categories mutually exclusive?
12. Give one business interpretation of .